

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.

- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id almr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Output – Screenshots:

Original Message

Message ID	<f9a50498-8499-44d4-91f7-eca0f8cea891@dfw1s10mta1942.xt.local>
Created at:	Sun, Aug 9, 2026 at 9:50 PM (Delivered after 1 second)
From:	edX Team <news@sfmc.edx.org>
To:	tejeshwarreddyilluri@gmail.com
Subject:	Celebrate IBM PC Day by learning what comes next
SPF:	PASS with IP 13.111.127.7 Learn more
DKIM:	'PASS' with domain sfmc.edx.org Learn more
DMARC:	'PASS' Learn more

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Email Header Analyzer

Paste Header:

```

</tbody></table>
</div>
=20
<img src=3D"https://d5ilC804.na1.hubspotlinks.com/Clo/DQ+113/d5ilC804/VVvx=
_7h9xV3W7SndZ78N3pnhW5j5c_V5StLBJW3cFvTP8fYG2531Xt1" alt=3D"" h=
eight=3D"1" border=3D"0" style=3D"display:none!important,min-height:1px!imp=
ortant,width:1px!important;border-width:0!important;margin-top:0!important;=
margin-bottom:0!important;margin-right:0!important;margin-left:0!important;=
padding-top:0!important;padding-bottom:0!important;padding-right:0!important=
padding-left:0!important"></body></html>
-----_Part_674963_825960495.1786235384878--

```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

MessageId	<f9a50498-8499-44d4-91f7-eca0f8cea891@dfw1s10mta1942.xt.local>
Created at:	8/9/2026, 9:50:11 PM GMT+5:30 (Delivered after 1 sec)
From:	"edX Team" <news@sfmc.edx.org>
To:	<tejeshwarreddyilluri@gmail.com>
Subject:	Celebrate IBM PC Day by learning what comes next
DKIM:	pass with domain sfmc.edx.org Learn more

#	Delay	From *	To *	Protocol	Time received
0	1 sec	mta2.sfm.edx.org.	→ [Google] mx.google.com	ESMTPS	8/9/2026, 9:50:12 PM GMT+5:30
1			→ [Google] 2002:a05:6870:880:bd0:448:3eb2:8cfc	SMTP	8/9/2026, 9:50:12 PM GMT+5:30
2			→ [Google] 2002:a05:7010:8b89:bc0:534:39cf:de51	SMTP	8/9/2026, 9:50:12 PM GMT+5:30

Whois IP 143.244.91.119

Updated 1 second

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      143.244.80.0 - 143.244.95.255
CIDR:          143.244.80.0/20
NetName:       HUBSP-8
NetHandle:     NET-143-244-80-0-1
Parent:        NET143 (NET-143-0-0-0-0)
NetType:       Direct Allocation
OriginAS:
Organization:  HubSpot, Inc. (HUBSP-8)
RegDate:       2019-12-27
Updated:       2021-04-21
Comment:       -----BEGIN CERTIFICATE-----MIIDSzCCApugAwIBAgIUTwbH/4z+Ax6hsCUGcjxxQrKOIOgwdQYJl
Ref:           https://rdap.arin.net/registry/ip/143.244.80.0

OrgName:       HubSpot, Inc.
OrgId:         HUBSP-8
Address:       2 Canal Park
City:          Cambridge
StateProv:     MA
PostalCode:    02141
Country:       US
RegDate:       2018-10-08
```

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result : The email header was successfully analyzed using Mail Header Analyzer. The Received fields, IP addresses, Message-ID, SPF, DKIM and DMARC information were examined to identify possible anomalies and detect email spoofing.